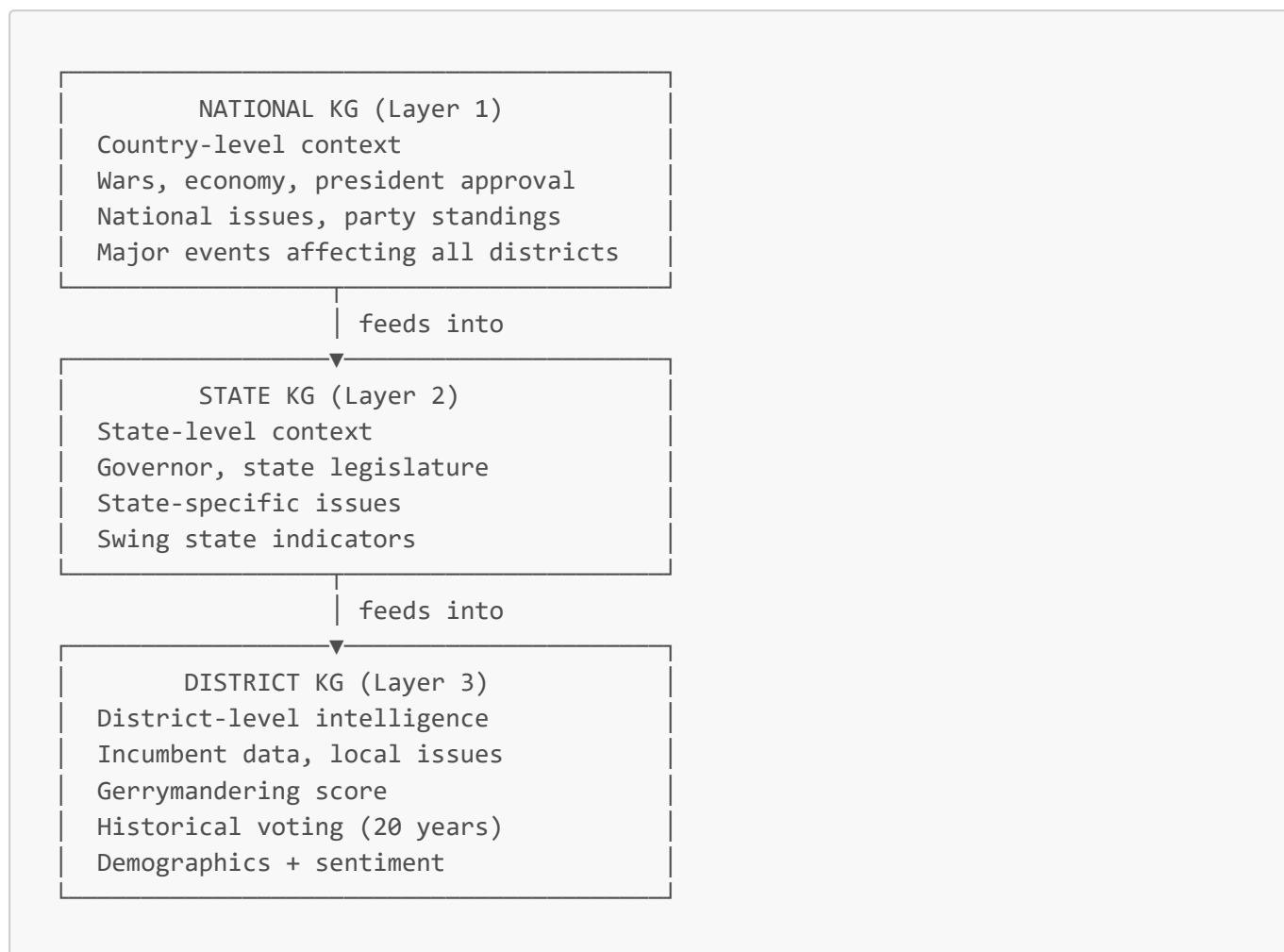


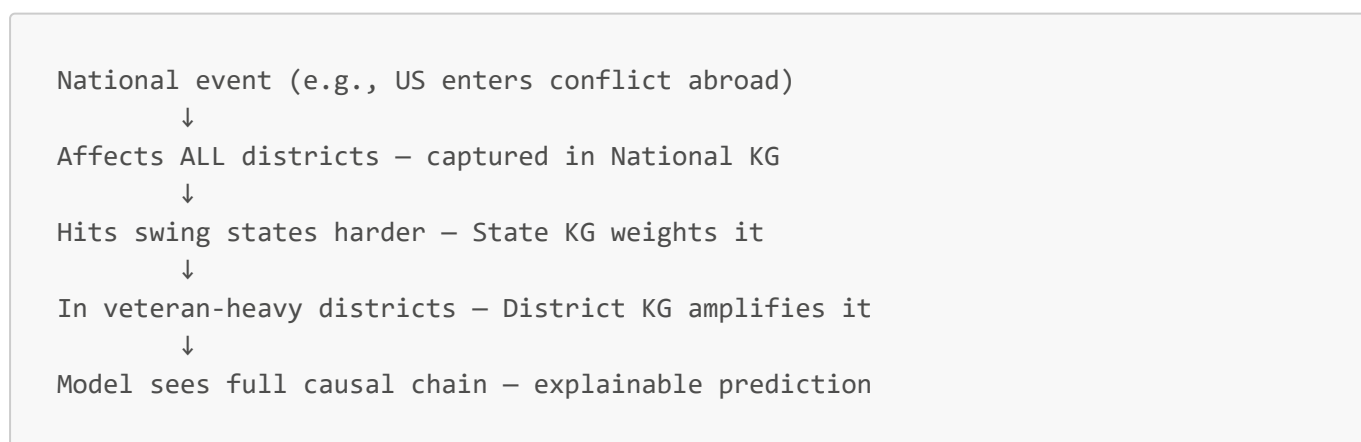
Data Sources, KG Architecture & Key Indicators

Political Intelligence Platform — Data Strategy Document

1. KG Architecture — Three-Layer Design



Why Three Layers?



This is your biggest architectural contribution. No existing public model explicitly models this cascade.

2. Key Static Indicators — Build These First

These are your foundational node properties and relationships. Every district KG node must have all of these before anything else.

2.1 Incumbent Indicators

Is there an incumbent running?	(boolean)
Incumbent party	(R/D/I)
Incumbent approval rating	(0-100)
Incumbent years in office	(integer)
Incumbent fundraising total (FEC)	(dollars)
Incumbent key votes (for/against)	(list)
Incumbent scandal flags	(boolean)
Challenger quality score	(weak/strong/wave)
Open seat?	(boolean – no incumbent)

Open seats are the most volatile — weight these heavily in your model. Incumbents win 90%+ of the time. Open seats flip 3x more often.

2.2 Swing Indicators

Cook Political Report rating	(Safe R → Toss-up → Safe D)
Sabato Crystal Ball rating	(same scale)
Dave's Redistricting competitiveness	(0-100 score)
Past 3 election margin average	(percentage)
Margin trend direction	(narrowing/widening)
Presidential vs midterm swing delta	(how much it changes)
Ticket splitting rate	(voters who split R/D)

2.3 Gerrymandering Indicators

PlanScore partisan bias score	(-1 to +1)
Princeton Gerrymandering grade	(A-F)
Last redistricting year	(2021/2022 most recent)
Who controlled redistricting	(R legislature/D legislature/court)
District compactness score	(geometric measure)
Population deviation %	(how equal are districts)
Legal challenges filed?	(boolean)

2.4 Historical Voting (20 Years)

Presidential results 2004-2024	(R%, D%, margin)
House results 2004-2024	(R%, D%, margin)
Midterm vs presidential turnout delta	(percentage)

Average R margin last 5 cycles	(percentage)
Trend direction	(redder/bluer/stable)
Wave election behavior	(did it flip in 2010/2018?)

2.5 War & Foreign Policy Indicators

Active conflict period flag	(boolean, which conflict)
Veteran population %	(Census)
Active military base in district	(boolean)
Defense contractor employment %	(Bureau of Labor Stats)
Gold Star family concentration	(proxy: veteran orgs density)
Casualty impact index	(if available)
Foreign policy salience score	(from news volume)

War effects are asymmetric:

- Veteran-heavy districts: foreign policy matters 3x more
- Urban districts: war matters less than economy Your KG captures this via district-demographic relationships

2.6 Economic Indicators

Unemployment rate	(BLS API)
Median household income	(Census)
Income growth/decline trend	(year over year)
Industry concentration	(manufacturing/tech/agriculture)
Union membership %	(BLS)
Cost of living index	(BLS CPI by metro)
Recent major layoffs	(news scraping)
Small business density	(Census Business Patterns)

2.7 Demographic Indicators

Total population	(Census)
Age distribution (18-34, 35-54, 55+)	(Census)
Race/ethnicity breakdown	(Census)
Education level (college grad %)	(Census)
Urban/suburban/rural classification	(Census)
Population growth rate	(Census)
Foreign born %	(Census)
Religious affiliation (proxy)	(PRRI data)

2.8 Issue Salience Indicators (Static Baseline)

Healthcare concern score	(from survey data)
Immigration concern score	(from survey data)
Economy/inflation concern score	(from survey data)
Education concern score	(from survey data)
Crime/safety concern score	(from survey data)
Climate concern score	(from survey data)
Gun policy concern score	(from survey data)

3. Static Phase — All Data Sources

3.1 Election & Political Data

Source	What to Scrape/Pull	Method	Free?
MIT Election Lab	House results 2000-2024 by district	Direct download CSV	✓
Harvard Dataverse	Historical election datasets	Direct download	✓
FEC API	Campaign finance by candidate/district	REST API	✓
Ballotpedia	Candidate bios, race ratings, incumbents	Web scrape + API	✓
Congress.gov API	Voting records, bills sponsored	REST API key (free)	✓
GovTrack	Legislator scores, voting records	Web scrape + data downloads	✓
VoteSmart	Candidate positions on issues	Web scrape	✓
OpenSecrets	PAC funding, dark money	Web scrape	✓
Wikipedia	Historical election results pages	Web scrape	✓

3.2 Swing / Competitiveness Ratings

Source	What to Pull	Method	Free?
Cook Political Report	Race ratings by district	Web scrape (HTML table)	✓
Sabato's Crystal Ball	Race ratings	Web scrape	✓
Dave's Redistricting	Competitiveness scores	Web scrape + data export	✓
PlanScore	Gerrymandering scores	API + web scrape	✓
Princeton Gerrymandering Project	District grades	Web scrape	✓
FiveThirtyEight data repo	Historical ratings + polls	GitHub CSV download	✓

```
# FiveThirtyEight data – direct download, no scraping needed
git clone https://github.com/fivethirtyeight/data
```

3.3 Demographics & Census

Source	What to Pull	Method	Free?
US Census ACS 5-Year	Demographics by district	REST API	✓
Census TIGER	District shapefiles/boundaries	Direct download	✓
Census Business Patterns	Industry employment	REST API	✓
BLS API	Unemployment by county	REST API	✓
BEA API	GDP, income by region	REST API	✓
PRRI	Religious landscape data	Web scrape	✓

```
# Census API example – get demographics for all congressional districts
import requests

url = "https://api.census.gov/data/2022/acs/acs5"
params = {
    "get": "NAME,B01003_001E,B19013_001E,B15003_022E,B02001_003E",
    # population, median income, bachelor's degree, Black population
    "for": "congressional district:*",
    "in": "state:*",
    "key": "YOUR_FREE_KEY" # get at api.census.gov/data/key_signup.html
}
data = requests.get(url, params=params).json()
```

3.4 War & Military Data

Source	What to Pull	Method	Free?
DoD Base Structure Report	Military bases by state/county	PDF download + parse	✓
VA Data	Veteran population by district	Direct download	✓
Defense Contract Database	Defense jobs by district	USASpending.gov API	✓
USASpending.gov API	Federal spending by district	REST API	✓
Casualties data	Historical war casualties by state	Wikipedia scrape	✓

3.5 News (Static Baseline — Historical)

Source	What to Pull	Method	Free?
GDELT 2.0	All political news 2015-present	BigQuery or direct API	✓
CommonCrawl	Full web archive	S3 download	✓
Internet Archive	Historical news pages	API	✓
NewsAPI (historical)	1 month free historical	REST API	Limited

Source	What to Pull	Method	Free?
MediaCloud	Political media tracking	API	✓

```
# GDELT – completely free, no key, massive coverage
import requests

url = "https://api.gdeltproject.org/api/v2/doc/doc"
params = {
    "query": "Arizona 4th congressional district healthcare 2024",
    "mode": "artlist",
    "maxrecords": 250,
    "startdatetime": "20240101000000",
    "enddatetime": "20241101000000",
    "format": "json"
}
articles = requests.get(url, params=params).json()
```

3.6 Social Media (Static Historical)

Source	What to Pull	Method	Free?
Pushshift Reddit	Historical Reddit posts by subreddit	API + dumps	✓
Reddit PRAW	Current + recent posts	Python library	✓
YouTube Data API	Political video comments	REST API key (free)	✓
Twitter Academic	Historical political tweets	Apply for academic access	✓

3.7 Polling Data

Source	What to Pull	Method	Free?
FiveThirtyEight polls	All historical polls by race	GitHub CSV	✓
RealClearPolitics	Poll averages by race	Web scrape	✓
Polling USA	District-level polls	Web scrape	✓

4. Dynamic Phase — Sources & Strategy

4.1 What Changes Weekly

Every week your pipelines should update:

National KG:

- └─ President approval rating (Gallup weekly)
- └─ Economic indicators (BLS monthly release)
- └─ Major national events (GDELT)

└─ Congressional votes (Congress.gov)

└─ National polling averages (FiveThirtyEight)

State KG (Swing States Priority):

└─ State polling (RCP)

└─ Governor approval

└─ State-level news volume by issue

└─ Fundraising updates (FEC quarterly)

District KG:

└─ Local news sentiment (GDELT filtered by district)

└─ Reddit discussion volume by issue

└─ Issue salience scores (updated from news)

└─ Candidate activity + announcements

4.2 Swing States — Focus List 2026

These get priority in your dynamic phase:

Arizona	Pennsylvania	Wisconsin
Georgia	Michigan	Nevada
North Carolina	New Hampshire	Minnesota
Florida	Ohio	Virginia

For these states: pull data **daily** not weekly. For safe states: weekly is enough.

4.3 Dynamic Data Sources

Source	Update Frequency	What It Gives	Method
GDELT 2.0	Every 15 minutes	All political news	API — no key
NewsAPI	Daily	Top news by query	REST API
Reddit PRAW	Daily	Issue discussions	Python library
FEC API	Quarterly	Fundraising updates	REST API
Congress.gov	Weekly	New bills and votes	REST API
BLS releases	Monthly	Jobs, inflation	REST API
Gallup	Weekly	Presidential approval	Web scrape
RealClearPolitics	Daily	Poll averages	Web scrape
FiveThirtyEight	Daily	Forecast updates	Web scrape + GitHub
Google Trends API	Daily	Issue search volume	pytrends library
YouTube Data API	Weekly	Political video comments	REST API

```
# Google Trends – free, shows what issues people are searching
from pytrends.request import TrendReq

pytrends = TrendReq()
pytrends.build_payload(
    ["healthcare", "inflation", "immigration", "crime"],
    geo="US-AZ",          # Arizona
    timeframe="today 3-m"
)
df = pytrends.interest_over_time()
# Returns weekly search volume per issue per state
# Feed directly into issue salience scores in your KG
```

```
# Airflow DAG – weekly update pipeline
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime, timedelta

def update_district_kg():
    # 1. Pull new GDELT articles
    # 2. Extract issues via NLP
    # 3. Update issue node weights in Neo4j
    # 4. Run PyKEEN link prediction
    # 5. Add inferred relations to KG
    pass

dag = DAG(
    "weekly_kg_update",
    schedule_interval="@weekly",
    start_date=datetime(2026, 6, 1)
)

update_task = PythonOperator(
    task_id="update_kg",
    python_callable=update_district_kg,
    dag=dag
)
```

4.4 Event Detection — What Triggers a KG Update

Not every piece of news warrants a KG update. Your pipeline should flag events that change district dynamics:

```
# Event types that trigger KG weight updates
TRIGGER_EVENTS = {
    "economic": [
        "unemployment rate", "inflation report",
        "factory closure", "major layoffs", "plant opening"
    ],
}
```



```

    "political": [
        "candidate drops out", "endorsement",
        "scandal", "debate", "ad campaign launch"
    ],
    "national": [
        "military action", "supreme court ruling",
        "presidential approval drop", "major legislation passed"
    ],
    "local": [
        "district-specific crime surge", "local disaster",
        "school shooting", "healthcare facility closure"
    ]
}

# When detected → update relevant issue node weight in KG
# Example: "factory closure in AZ-07" →
#         Economy issue node weight += 0.15 in AZ-07 district KG

```

5. Final KG Schema With All Indicators

```

// National level node
CREATE (:National {
    president_approval: 42.3,
    gdp_growth: 2.1,
    inflation_rate: 3.4,
    active_conflicts: ["Ukraine support", "Middle East"],
    national_issue_top3: ["Economy", "Immigration", "Healthcare"],
    updated_at: datetime()
})

// State node
CREATE (:State {
    name: "Arizona",
    code: "AZ",
    governor: "Katie Hobbs",
    governor_party: "Democrat",
    governor_approval: 48.2,
    is_swing_state: true,
    electoral_votes: 11,
    state_legislature_control: "Split",
    last_presidential_result: "Democrat +0.3",
    updated_at: datetime()
})

// District node – full indicators
CREATE (:District {
    id: "AZ-06",
    name: "Arizona 6th Congressional District",
    state: "AZ",

```

```
// Incumbent
incumbent_name: "Juan Ciscomani",
incumbent_party: "Republican",
incumbent_approval: 44.1,
incumbent_years: 2,
incumbent_fundraising: 2100000,
open_seat: false,
incumbent_scandal: false,

// Swing indicators
cook_rating: "Toss-up",
sabato_rating: "Toss-up",
competitiveness_score: 0.89,
avg_margin_3cycles: 1.2,
margin_trend: "narrowing",

// Gerrymandering
gerrymander_score: 0.12,
princeton_grade: "B",
redistricted_year: 2022,
redistricting_control: "Republican legislature",
district_compactness: 0.61,

// Historical voting
presidential_2020: "Democrat +2.1",
presidential_2016: "Republican +3.4",
presidential_2012: "Republican +8.2",
house_2022: "Republican +0.8",
house_2020: "Republican +5.2",
avg_r_margin_5cycles: 3.9,
trend_direction: "bluer",

// Demographics
population: 761169,
median_income: 68420,
college_grad_pct: 34.2,
hispanic_pct: 38.1,
white_pct: 51.3,
age_median: 38.4,
urban_pct: 72.1,
veteran_pct: 9.8,

// Military/War
military_base_present: false,
defense_employment_pct: 4.2,
veteran_population: 74594,

// Economic
unemployment_rate: 4.1,
income_trend: "declining",
major_industries: ["Healthcare", "Retail", "Construction"],
union_membership_pct: 8.3,

updated_at: datetime()
```

```

}))

// Issue nodes
CREATE (:Issue {
  name: "Healthcare",
  category: "Domestic",
  national_salience: 72.3,    // % of people who care nationally
  updated_at: datetime()
}))

// Key relationships
MATCH (d:District {id: "AZ-06"}), (i:Issue {name: "Healthcare"})
CREATE (d)-[:CONCERNED_ABOUT {
  weight: 0.78,                // how much this district cares
  trend: "increasing",         // growing or shrinking concern
  source: "GDELT+Reddit",
  sentiment: -0.34,            // negative = unhappy with current state
  updated_at: datetime()
}]->(i)

```

6. What Your Model Actually Predicts

Not just: "District X → Republican wins"

But:

"District X – Toss-up leaning Republican (54%)"

- Reason 1: Incumbent has 2yr advantage (KG: incumbent node)
- Reason 2: But margin narrowing – was +8 now +0.8 (KG: historical)
- Reason 3: Healthcare concern surged 34% (KG: issue weight ↑)
- Reason 4: Incumbent voted against ACA expansion (KG: vote relation)
- Reason 5: Similar to FL-26 in 2018 – flipped Democrat (KG: analogy)
- Risk flag: HIGH – issue dynamics moving against incumbent

That's a 538 prediction + Cook rating + your KG reasoning chain. All three together. That's your contribution.

7. Build Priority Order

- Week 1-2: Census + FEC + MIT Election Lab → PostgreSQL
Build district nodes with all static indicators
- Week 3-4: Historical voting 2004-2024 → KG relationships
Gerrymandering scores → PlanScore + Princeton scrape
Incumbent data → Ballotpedia scrape
- Week 5-6: GDELT static pull (2024 news) → issue extraction
Baseline NLP: sentiment + issue detection per district
XGBoost baseline model with all indicators

Week 7-8: Full static KG built for 50 key districts
Neo4j visualization working
Basic Streamlit dashboard

Week 9-10: Add Airflow → weekly GDELT + Reddit pipeline
Dynamic KG weight updates
Google Trends integration

Week 11+: Swing state focus – daily updates
PyKEEN link prediction layer
API + full dashboard

Data Strategy Document v1.0 — May 2026